Lecture 7

DAGs, Bad Controls, and Sensitivity Analysis

ECON6083: Machine Learning in Economics

Today's Roadmap

#	TOPIC	FOCUS
1	Modern Identification	Bridging structure and design
2	DAGs	Four elemental structures
3	Bad Controls	M-bias, collider bias
4	Sensitivity Analysis	Bounds, robustness, unobserved confounding
5	Causal Discovery	PC algorithm, limits, faithfulness
6	Applied Workflow	DAG first, DML second

Part 1

The Modern Identification Synthesis, DAGs, and SCMs

The Historical Divide

Period	1940s–1970s (Cowles Commission)	1990s–present
Approach	Deep structural parameters, decision models	Natural experiments, IV, RD
Promise	Counterfactual policy analysis	Credible, design-based estimates
Cost	Strong parametric assumptions	Limited external validity
Key figures	Haavelmo, Koopmans, Lucas	Angrist, Card, Imbens, Krueger

The New Synthesis (*JEP* 2010 Symposium)

c. 2010s – present: combine credibility of experiments with counterfactual power of models

- **Nevo & Whinston (2010)**: Both sides have dogma — good work needs credible ID *and* structural power
- **Einav & Levin (2010)**: Modern IO already does this — IV estimates → structural simulations
- **Keane (2010)**: Without a model, you cannot answer GE questions or extrapolate
- **Chetty (2009)**: Sufficient statistics offer a middle ground — estimate elasticities, plug into formulas

Key insight: No longer forced to choose between "clean design" and "structural mechanisms."

The Three Pillars and Today's Toolkit

1. Rigorous causal language — SCMs and DAGs

2. Flexible estimation — Double-Debiased ML

3. Theoretical foundations — Structural models for counterfactuals

Today: Draw DAGs → Audit controls → Sensitivity analysis → DML

DAG approach: Explicit causal flow via graphs

- **Nodes** = Variables; **Directed edges** = Causal influence; **Acyclic** = No feedback loops
- Think of paths as **pipes**: association flows unless "blocked" at a collider or conditioned fork

Part 2

The Four Elemental Structures

What Does "Open" vs. "Blocked" Mean?

A **path** is any sequence of edges connecting two variables.

- **OPEN path** = Statistical association can flow through it. Two variables on an open path may appear correlated, even if there is no direct causal link.
- **BLOCKED path** = The flow of association is stopped. Two variables on a blocked path are (conditionally) independent.

The core question for each structure:

> *"If we put this variable in our regression (condition on it), does the path become OPEN or BLOCKED? And is that good or bad for identifying the causal effect of X on Y?*

Structure 1: The Fork (Common Cause)

Configuration: $Z \to X$ and $Z \to Y$

Example: Age → Income and Health

What it does: Z creates a spurious correlation between X and Y.

ASPECT	STATUS
Path status (unconditioned)	**OPEN** — association flows along $X \leftarrow Z \to Y$
Effect of controlling for Z	**BLOCKS** the path
Verdict	**INCLUDE** ✓

> *We want to block this non-causal path. Conditioning on Z shuts off spurious correlation and isolates the true effect of X on Y.*

Structure 2: The Chain (Mediator)

Configuration: $X \to M \to Y$

Example: Education → Skills → Wages

What it does: M transmits part or all of the causal effect from X to Y.

ASPECT	STATUS
Path status (unconditioned)	**OPEN** — this is the causal mechanism itself
Effect of controlling for M	**BLOCKS** the path
Verdict	**EXCLUDE** × (for **total effect**)

> *Only control for M if estimating the **direct effect** or doing **mediation analysis**. Otherwise it's **over-control**.*

Structure 3: The Collider (Common Effect)

Configuration: $X \to C \leftarrow Y$

Example: Study Hours → College Admission ← Test Prep Quality

ASPECT	STATUS
Path status (unconditioned)	**BLOCKED**
Effect of controlling for C	**OPENS** the path
Verdict	**EXCLUDE** ×

> *"In the general population, X and Y are independent. But condition on C and they suddenly look correlated. This is the **collider bias trap**.*

Collider: The Battery-Lightbulb Intuition

Setup: Battery (X) and Lightbulb (Y) both needed for Light (C)

General population: $X \perp Y$ (battery and bulb are independent)

Conditioning on "Light is OFF" ($C = 0$):

- If battery works ($X = 1$) → bulb must be broken ($Y = 0$)
- Creates a *negative* correlation between two otherwise independent things!

Lesson: Conditioning on a collider is the structural foundation of **selection bias**. Do not control for common effects.

Structure 4: The Descendant (Child of Collider)

Configuration: $X \to C \leftarrow Y$ and $C \to D$

Example: Ability → Job Offer ← Connections; Job Offer → Income

What it does: D is a "noisy proxy" for the collider C.

ASPECT	STATUS
Path status (unconditioned)	**BLOCKED** — naturally closed
Effect of controlling for D	**PARTIALLY OPENS** the path
Verdict	**EXCLUDE** ×

> ***Why?*** *Controlling for D gives partial information about C, inadvertently leaking collider bias (e.g., controlling for income when income descends from an unobserved collider).*

Summary: The Four Structures

STRUCTURE	PATH STATUS	IF YOU CONDITION ON IT	ACTION
Fork	**OPEN** (spurious)	**BLOCKS** path	**Include** ✓
Chain	**OPEN** (causal)	**BLOCKS** path	**Exclude** ✕
Collider	**BLOCKED**	**OPENS** path	**Exclude** ✕
Descendant	**BLOCKED**	**Partially opens**	**Exclude** ✕

Part 3

The Taxonomy of "Bad Controls"

Beyond "Kitchen Sink" Regression

Traditional econometric intuition:

- Control for everything available to minimize omitted variable bias
- More controls = better identification
- "Kitchen sink" approach

DAG theory reveals: This is formally incorrect!

Critical distinction needed:

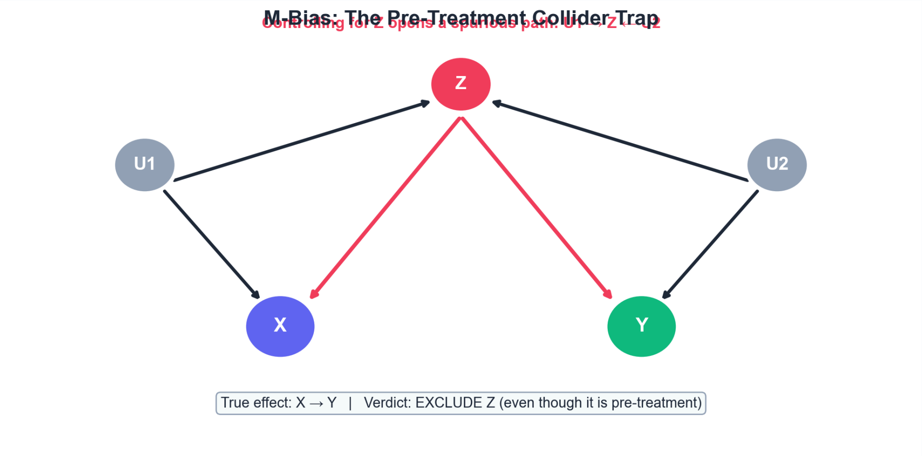
- **Good controls**: Confounders that close backdoor paths (forks)
- **Bad controls**: Variables that introduce or amplify bias
    - Mediators (block causal mechanisms)
    - Colliders (open spurious paths)
    - Descendants of colliders (noisy collider bias)

Bad Control 1: M-Bias

Source: Pearl (2009) *Causality*; Cinelli, Forney & Pearl (2022) *A Crash Course in Good and Bad Controls*

DAG: $U_1 \to X, U_1 \to Z \leftarrow U_2, U_2 \to Y$

STATE	RESULT
Without Z	Path blocked at collider Z → **no bias**
Controlling for Z	Path opens, spurious bridge U_1–U_2 → **bias**
Verdict	**EXCLUDE** Z; pre-treatment status alone is insufficient

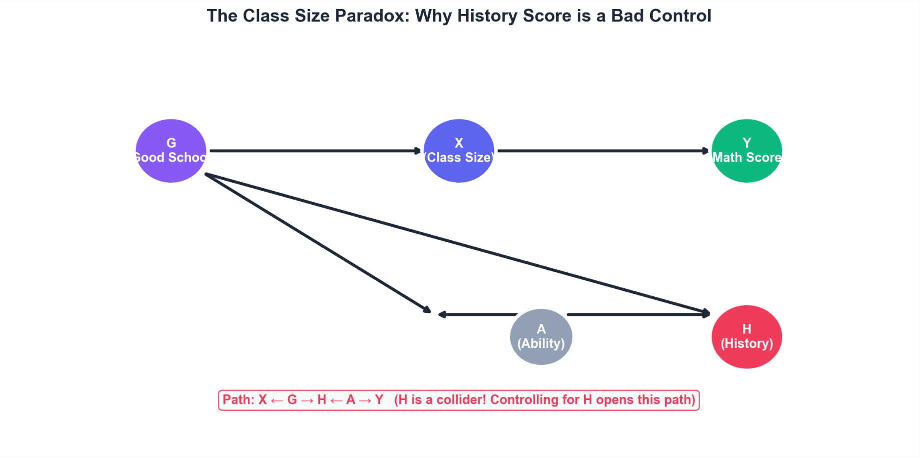


Bad Control 2: The Class Size Paradox

Source: Stylized example from Pearl (2009) DAG pedagogy

Setup: Effect of Class Size (X) on Math Score (Y)

PATH	ACTION
Path 1: $X \leftarrow G \rightarrow Y$ (confounding)	Need to **block** — include G
Path 2: $X \leftarrow G \rightarrow H \leftarrow A \rightarrow Y$ (collider H)	**Leave alone**
Controlling for H	Opens Path 2 → **bias**
Verdict	Include G (school quality); **EXCLUDE** H (History Score)

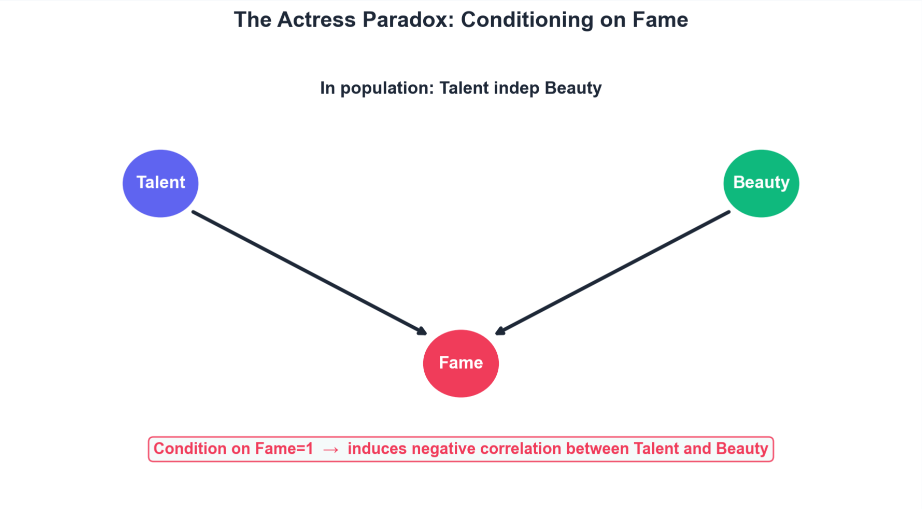


Bad Control 3a: The Actress Paradox

Source: Pearl & Mackenzie (2018) *The Book of Why*

DAG: $Talent \to Fame \leftarrow Beauty$

STATE	RESULT
Population	Talent and Beauty are independent
Condition on Fame (=1)	Low talent implies high beauty → **negative correlation**
Verdict	Do not condition on colliders (sample selection)

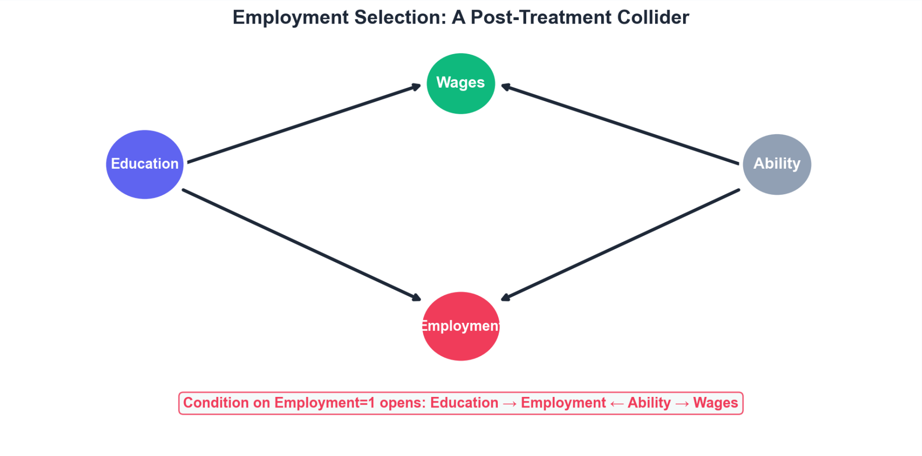


Bad Control 3b: Employment Selection

Source: Heckman (1979); Elwert & Winship (2014)

DAG: $Education \to Employment \leftarrow Ability \to Wages$

STATE	RESULT
Condition on Employment (=1)	Opens collider path → high-education, marginally employed implies lower ability → **downward bias**
Verdict	**Exclude** employment; use Heckman correction or selection model



Part 4

Sensitivity Analysis & Bounds

The Unobserved Confounder Problem

The dream: DAGs say, "Block all backdoor paths and the causal effect is identified."

Reality: There are almost always variables we cannot observe.

Classic example: Returns to Schooling

- You control for parents' income, school quality, location...
- But you do **not** observe **Ability**.
- Smarter people study more *and* earn more regardless of schooling.
- So part of the "schooling effect" you estimate is really picking up ability.

The key question: How strong would this unobserved Ability need to be to completely overturn my conclusion?

Oster (2019): A Simple Intuition

Oster's question: Given all observed controls, how strong would **unobserved Ability U** need to be to push the returns-to-schooling estimate from 8% down to 0%?

She answers this with three numbers:

SYMBOL	MEANING	EXAMPLE
$\tilde{\beta}_{short}$	Coefficient without controls	Returns = 15%
$\tilde{\beta}_{long}$	Coefficient with all observed controls	Returns = 8%
$\tilde{R}^2_{long}$	R^2 of the long regression	e.g., $R^2 = 0.25$

Key assumption: Proportional Selection

> *"The ratio of U's influence on schooling vs. wages equals the same ratio for the observed controls.*

Oster's δ: A One-Number Robustness Check

Under this assumption, Oster boils everything down to one parameter: δ **(delta)**

What δ means: The strength of the unobserved confounder U **relative to the observed controls**.

δ	MEANING	VERDICT
$\delta = 0.1$	Ability 1/10 as strong as observed controls suffices to erase result	**Very fragile**
$\delta = 1.0$	Ability must equal strength of observed controls to overturn	**Reasonably robust** ✓
$\delta = 2.0$	Ability must be twice as strong as observed controls	**Very robust** ✓✓

> ***Rule of thumb:*** *$\delta > 1$ is comforting. $\delta < 0.5$ should make you nervous.*

Cinelli & Hazlett (2020): A Better Picture

Critiques of Oster:

1. **Proportional selection is very strong**
2. **R^2 is not rescaling-invariant**

Improvement: Use Partial R^2

> *What fraction of the **residual variance** in Y and in X must U explain?*

Output: Contour plot

- Horizontal axis: U's effect on X (partial R^2)
- Vertical axis: U's effect on Y (partial R^2)
- Curve: Combinations that would push the coefficient to zero

> *If the curve lies outside (0.2, 0.2), U would need to explain >20% of residual variance in both — hard, so the result is **robust**.*

Manski Bounds: What If We Assume Nothing?

The other extreme: Make *zero* assumptions about the unobserved confounder.

The idea: I do not know what the unobserved confounder is or how strong it is. So I consider the **most pessimistic scenarios**.

Example *(hypothetical, for illustration)*:

- Average wage for college graduates = $15,000
- Average wage for non-graduates = $10,000
- But some non-graduates would have earned $30,000 if they had gone to college, while others would have earned only $5,000

Manski's approach: Bound the true causal effect by considering the best-case and worst-case counterfactuals for the control group.

Manski Bounds: A Numerical Walkthrough

Setup: College graduates earn $15,000 on average. Non-graduates earn $10,000. What is the true causal effect of going to college?

We never observe the counterfactual (what non-graduates would have earned if they had gone to college). But we know wages must be between **$0** and **$30,000**.

SCENARIO	COUNTERFACTUAL WAGE (IF NO COLLEGE)	CAUSAL EFFECT = $15,000 − COUNTERFACTUAL
Most optimistic	$0	**$15,000**
Most pessimistic	$30,000	**−$15,000**

Manski bound: The true effect must lie somewhere in **[−$15,000, $15,000]**.

> *If we also bound the treatment group's counterfactual (graduates if they had not gone to college), the interval tightens further. The narrower the bound, the more confident we can be even with zero assumptions.*

Comparing Sensitivity Approaches

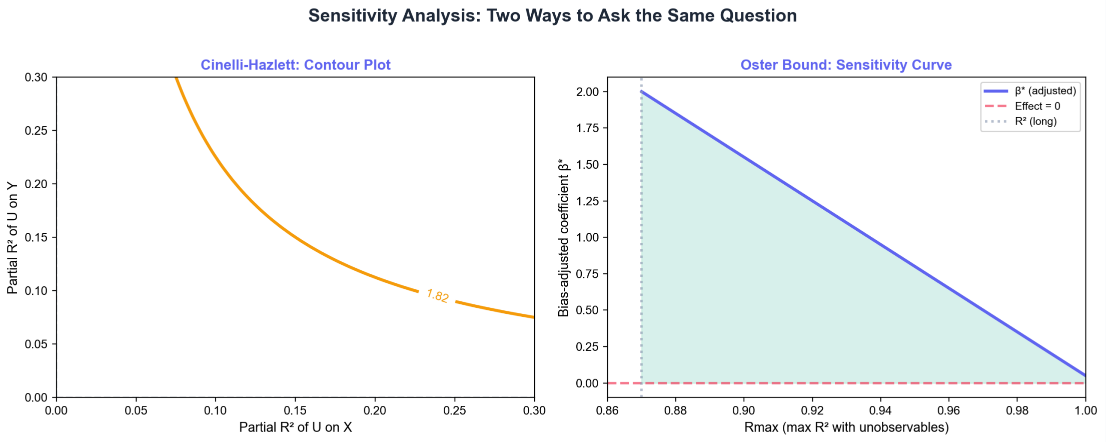
METHOD	ASSUMPTION	OUTPUT	WHEN TO USE
Oster (2019)	Proportional selection	Scalar δ	Quick robustness check
Cinelli-Hazlett	None (partial R^2)	Contour plot	Formal sensitivity analysis
Manski	None	Interval bounds	Extreme skepticism
DAG-based	Full causal structure	Include/exclude	Variable selection

Sensitivity Analysis in Pictures

Left: Cinelli-Hazlett contour plot shows how strong an unobserved confounder would need to be (in terms of partial R^2 on X and Y) to reduce your estimate to specific values.

Right: Oster bound traces how the bias-adjusted coefficient changes as you allow the unobserved confounder to explain more residual variance (Rmax).

> *Both ask the same question: How strong would a missing confounder need to be to overturn my result?*



Part 5

Causal Discovery

Can We Learn DAGs from Data?

Causal Discovery: Algorithms that attempt to learn causal structure from observational data.

Why it's appealing:

- Don't need prior theory?
- Let the data "speak"?
- Scale to high dimensions?

Reality: Partially possible, but with strong assumptions and serious limitations.

The PC Algorithm: The Big Picture

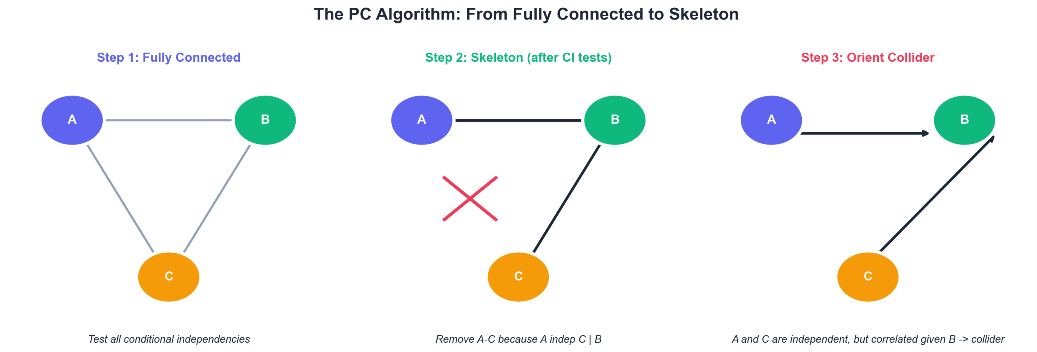
Goal: Learn the DAG from conditional independence (CI) tests in the data.

Why this is hard: Data only tells you *correlations* and *independencies*. You must infer *causal structure* from them.

Three-step logic:

1. **Skeleton**: Find which variables are directly connected.
2. **Orientation**: Use collider patterns to figure out which way arrows point.
3. **Propagation**: Apply logical rules (no cycles, no new V-structures) to fill in the rest.

Key assumption: **Faithfulness** — every d-separation in the true DAG shows up as a conditional independence in the data.



Step 1: Skeleton Phase

Start: Fully connected graph (every pair of variables has an edge).

Test: For each pair (X, Y), check if $X \perp Y \mid Z$ for some conditioning set Z.

Rule: If X and Y are conditionally independent, **remove the edge**.

Intuition: If X and Y become independent after controlling for other variables, there is no direct causal link between them.

Example: In $A \to B \to C$, A and C are marginally correlated, but $A \perp C \mid B$. PC removes the $A - C$ edge.

Step 2: Orientation Phase — Detecting Colliders

Setup: In the skeleton, we see $X - Z - Y$ but **no edge** between X and Y.

Question: Is Z a mediator ($X \to Z \to Y$), a fork ($Z \to X, Z \to Y$), or a collider ($X \to Z \leftarrow Y$)?

Collider test: Find a set S (not containing Z) such that:

- $X \perp Y \mid S$ (conditioning on S blocks the path)
- But $X \not\perp Y \mid S \cup \{Z\}$ (adding Z **opens** the path)

Why this works: Only a collider has the property that conditioning on it *creates* rather than *blocks* association.

Result: Orient as $X \to Z \leftarrow Y$.

Step 3: Propagation Rules

Once some arrows are oriented, we can deduce the rest using two logical constraints:

Rule 1 — No cycles

> *If $A \to B$ and $B \to C$ are already oriented, then $A - C$ must be $A \to C$ (not $C \to A$), otherwise we would create a cycle.*

Rule 2 — No new unshielded colliders

> *If $A \to B - C$ and there is **no edge** between A and C, then $B - C$ must be $B \to C$. If it were $C \to B$, we would create an undetected collider $A \to B \leftarrow C$.*

Iterate until no more arrows can be oriented.

Why Faithfulness Matters

Faithfulness: The data's conditional independencies *exactly* match the DAG's d-separations.

Why it is strong:

- **Path cancellation**: Two open paths with opposite effects can cancel out, making variables look independent even when connected.
- **Weak associations**: Small sample + noisy data = spurious independence.
- **Non-linearity**: PC typically uses linear correlation tests, which miss non-linear conditional independencies.

The domino effect: One wrong edge removal in the skeleton phase cascades into many wrong orientations later.

Limits and Uses of Causal Discovery

Limits:

CHALLENGE	PROBLEM
Hidden confounders	Cannot distinguish $X \to Y$ from $X \leftarrow U \to Y$
Markov equivalence	Different DAGs share the same conditional independencies
Sample size	CI tests grow exponentially; finite-sample errors cascade

When it helps: High-dim settings (gene networks), generating candidate DAGs, sensitivity checks across Markov-equivalent graphs

When it does not: Replacing theory, resolving endogeneity, justifying bad controls

> *Causal discovery is an **exploratory tool**, not a substitute for domain knowledge.*

The Right Role for ML in Causal Inference

TASK	ML TOOL	HUMAN ROLE
Prediction/forecasting	Random Forests, Neural Nets	Define target
High-dim control	DML (Lecture 5)	Choose DAG/controls
Heterogeneous effects	Causal Forests (Lecture 6)	Define subgroups
Sensitivity analysis	`sensemakr`, `bounds`	Interpret robustness
Causal discovery	PC, GES, NOTEARS	Validate structure

Core lesson: ML extends our **estimation** toolkit; it does not replace **identification** reasoning.

Part 6

Paper Applications

Cinelli, Forney & Pearl (2022): Good and Bad Controls

"A Crash Course in Good and Bad Controls"

Sociological Methods & Research, 2022

Research question: Which control variables reduce bias, and which introduce or amplify it?

ML: DAG-based structural taxonomy replacing ad hoc "more controls is better" heuristics

Key finding: Classification depends on structural position of Z in the DAG, not on statistical associations.

The Cinelli-Forney-Pearl Taxonomy

CATEGORY	EFFECT OF CONDITIONING ON Z	RECOMMENDATION
Good Controls	Reduces bias, may increase variance	**Always include**
Bad Controls	Introduces or amplifies bias	**Never include**
Neutral Controls	No effect on bias (may affect precision)	**Optional**

Classification depends on structural position of Z relative to $X \to Y$, not statistical associations.

Type 6-12: Neutral and Context-Dependent Controls

TYPE	STRUCTURE	EFFECT
Precision Variables	$Z \to Y$, not X	Increases precision, no bias effect
Competing Exposure	$X \leftarrow Z \to Y$ (instrument-like)	Neutral for bias, may help precision
Descendant of Good Control	Z caused by confounder	Generally safe, slightly less effective
Descendant of Collider	Z caused by collider	**BAD** — same as collider

Rule of thumb: When in doubt, draw the DAG and trace paths

Practical Decision Flowchart

Step 1: Is Z pre-treatment or post-treatment?

- **Post-treatment**: Only include if studying *direct* effects

Step 2: Is Z a common cause of X and Y?

- **Yes**: Include (confounder)

Step 3: Is Z caused by X or Y (or their ancestors)?

- **Yes**: Exclude (collider)

Step 4: Is Z caused by unobserved confounders of X and Y?

- **Yes**: Exclude (M-bias)

Elwert & Winship (2014): Endogenous Selection

"Endogenous Selection Bias: The Problem of Conditioning on a Collider Variable"
Annual Review of Sociology, 2014

Research question: Why does conditioning on endogenous variables create bias? How do DAGs unify confounding and selection bias?

Method: DAG collider analysis applied to observational sociology

Key finding: Selection bias is bias created by conditioning on a collider, not by bad sampling.

The Marriage Wage Premium: A Classic Illusion

The fact: Married men earn 10%-40% more than unmarried men.

Three possible explanations:

1. **Causal**: Marriage makes men more responsible, so employers pay them more.
2. **Selection**: Better men are more likely to marry (and more likely to earn more anyway).
3. **Collider trap** (most subtle): By looking *only* at married men, you **artificially open a spurious path**.

Elwert & Winship's key insight: The third explanation is not "bad sampling." It is **conditioning itself that creates the bias**.

Why Marriage Is a Collider

DAG: Marriage is not random. It depends on many things:

```
Economic Prospects ⟶ Marriage ⟵ Personal Charm
      ^                 ^             v
      |________ Wages <_______________|
```

- **Economic prospects** → higher wages, and more likely to marry
- **Personal charm** → higher wages (through social skills), and more likely to marry

In the whole population, economic prospects and personal charm might be independent.

But among married men: Low economic prospects + married implies especially charming → artificial negative correlation between prospects and charm → **contaminates the marriage effect estimate**.

The Peer-Effect Collider Trap

Research question: Do friends really influence each other's smoking behavior?

Reality of network data: We only observe friendships that *already formed*.

```
A's Smoking ⟶ Friendship ⟵ B's Smoking
     ^              ^               ^
     |_____ Similarity (U) ________|
```

The problem: When you restrict your sample to pairs who are friends, you are conditioning on a collider (friendship).

The result: Even with zero peer influence, A and B look similar — similar people become friends first. This is **homophily masquerading as peer influence**.

Four Strategies to Deal with Endogenous Selection

> *"Selection bias is not bad sampling. It is bias from conditioning on a collider.*

1. Sensitivity analysis — How strong would an unobserved confounder need to be? (Oster / Cinelli-Hazlett)

2. Selection models — Heckman correction: model *why this person is in my sample*.

3. Design-based fixes — IV or randomized interventions break the collider structure.

4. Manski bounds — Report the range of effects under worst-case counterfactuals.

Remember: Colliders are everywhere — marriage, friendship, employment. Draw the DAG first.

Greenland, Pearl & Robins (1999): Confounding Defined

"Causal Diagrams for Epidemiologic Research"
Epidemiology, 10(1), 1999

Research question: What is confounding, and when does stratification identify causal effects?

Method: DAG formalization of confounding and the backdoor criterion

Key finding: Confounding is structural, not statistical — "association with X and Y" is neither sufficient nor necessary:

- IVs are associated with both but are not confounders
- M-bias variables are unassociated but harmful if controlled

Why "Association" Is Not Enough to Define Confounding

Counterexample 1: Instrumental variable

- Rain (Z) predicts umbrella (X) and is associated with getting wet (Y) through X
- But controlling for Z **destroys identification**!

Counterexample 2: M-bias

- $U_1 \to Z \leftarrow U_2$, Z unrelated to both X and Y marginally
- Yet controlling for Z **creates bias**

GPR's formal definition: C confounds $X \to Y$ iff it is a cause of Y, not on the causal path, and opens a backdoor path $X \leftarrow C \to Y$

Confounding vs. Selection Bias: Finally Distinguished

Before GPR, these two concepts were often conflated.

Source	Pre-treatment common causes	Post-treatment conditioning / sample selection
Mechanism	Opens a backdoor path	Opens a collider path
Example	SES affects both smoking and cancer	Loss to follow-up in a clinical trial

Common feature: Both create non-causal associations between X and Y.
Key difference: One comes from a **pre-treatment** confounder, the other from **post-treatment** selection on a collider.

The Evolution of Causal Inference

PAPER	CORE CONTRIBUTION
Greenland et al. (1999)	Formal DAG-based definition of confounding
Elwert & Winship (2014)	Selection bias = conditioning on colliders
Cinelli et al. (2022)	Operational taxonomy of good/bad/neutral controls

Three legacies of GPR (1999):

1. "Association" → "causal effect": significance does not imply causation
2. "Control for variables" → "block backdoor paths"
3. "Vague bias" → specific structural mechanisms: forks, chains, colliders

How to Run a Project: The Applied Workflow

Step 1: Draw the DAG *before* touching the data — identify confounders, colliders, mediators

Step 2: Audit controls using Cinelli's taxonomy — include/exclude each Z, never kitchen-sink

Step 3: Check for selection bias — is your sample conditioned on a collider? Any attrition?

Step 4: Run sensitivity analysis — Oster's δ or Cinelli-Hazlett contours

Step 5: Estimate with DML — only after the DAG is correct

> ***Golden rule**: DML fixes estimation; it does **NOT** fix identification. If the DAG is wrong, DML will not save you.*

The Four Most Common DAG Mistakes

1. Drawing the DAG after seeing the data

> *Data-driven DAGs create circular reasoning. DAGs must come from theory.*

2. Conditioning on colliders

> *Opens spurious paths and introduces bias. One of the most common mistakes.*

3. Conditioning on post-treatment variables

> *Blocks legitimate causal pathways and leads to over-control.*

4. Ignoring unobserved confounding

> *DAGs can only handle the variables you draw. If important confounders are missing, you need IV, RDD, or bounds.*

Further Reading

READING	BEST FOR	CORE VALUE
The Book of Why (Pearl & Mackenzie, 2018)	Beginners	DAGs through stories
Causal Inference: The Mixtape (Cunningham, 2021)	Applied researchers	Code and examples
Causal Inference: What If (Hernan & Robins, 2020)	Advanced learners	Most comprehensive textbook

Further Reading (continued)

READING	BEST FOR	CORE VALUE
Cinelli et al. (2022), *AER Insights*	Everyone	Practical manual for control selection
Oster (2019), *JBES*	Applied researchers	Sensitivity analysis with R^2
Cinelli & Hazlett (2020), *JRSS-B*	Advanced learners	Partial R^2 contour plots

Data and Code

This lecture:

- Slides source: `_slides/Lecture07-DAGs and Structural Causal Models-Slides.md`
- Exercises: `exercises/Lecture07-Exercise.ipynb`

Key packages:

- Python: `dagitty`, `ggdag` (R), `networkx`

Datasets:

- IHDP (Infant Health Development Program)

Thank You!

Next Lecture: Instrumental Variables and DML-IV

Reading: Chernozhukov et al. (2018), Section IV